

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. STA/STB-502: Statistical Inference and Decision Theory

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) Define an unbiased estimator. What are the criteria of a good estimator? Show that $2\bar{X}$ is an unbiased estimator of θ in the rectangular distribution $U(0, \theta)$. [7]
- (b) State and prove the invariance property of a consistent estimator. Let $X_1, X_2, \dots, X_n$ be a random sample from a Bernoulli population with parameter θ . Show that $\bar{X}(1 - \bar{X})$ is a consistent estimator of $\theta(1 - \theta)$. [7]

Question 2.

- (a) Define a sufficient statistic. Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a Poisson distribution with parameter θ . Show that $\sum_{i=1}^n X_i$ is a complete sufficient statistic for θ . [7]
- (b) Let $X_1, X_2, \dots, X_n$ be a random sample from the probability density function:

$$f(x, \theta) = \frac{2x}{\theta^2}, \quad 0 < x < \theta, \quad \theta > 0.$$

Find the minimum variance unbiased estimator (MVUE) of θ . [7]

Question 3.

- (a) Stating the regularity conditions, state and prove the Cramér–Rao inequality. [7]
- (b) Let $X_1, X_2, \dots, X_n$ be a random sample from the probability density function:

$$f(x, \theta) = \theta e^{-\theta x}, \quad x \geq 0, \quad \theta > 0.$$

Obtain the minimum variance bound (MVB) estimator of θ . [7]

Question 4.

- (a) Explain the method of maximum likelihood estimation. Obtain the maximum likelihood estimator (MLE) of θ based on a random sample of size n from the population with probability density function:

$$f(x, \theta) = \frac{1}{\theta} x^{(1/\theta)-1}, \quad 0 < x < 1, \quad \theta > 0.$$

[7]

- (b) Let $X_1, X_2, \dots, X_n$ be a random sample from the probability density function:

$$f(x, \theta) = \frac{\theta^x e^{-\theta}}{x!(1 - e^{-\theta})}, \quad x = 1, 2, 3, \dots$$

Find the method of moments estimator of θ .

[7]

Question 5.

- (a) Explain the method of construction of confidence intervals. [4]
- (b) Obtain the $100(1 - \alpha)\%$ confidence interval for σ^2 of the normal distribution $N(\mu, \sigma^2)$ when:
- (i) μ is known,
 - (ii) μ is unknown.

[10]

Question 6.

- (a) Explain the following terms:
- (i) Simple and composite hypothesis
 - (ii) Type I and Type II errors

[7]

- (b) State and prove the Neyman–Pearson fundamental lemma.

[7]

Question 7.

- (a) Explain the likelihood ratio test. Obtain the likelihood ratio test for testing $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$ on the basis of a random sample of size n from $N(\mu, \sigma^2)$ with unknown σ^2 . [7]
- (b) Define the loss function and risk function. On the basis of a random sample of size n from a Poisson distribution with parameter θ , obtain the Bayes estimator of θ under the squared error loss function, when the prior distribution for θ is a Gamma distribution with parameters a and b . [7]

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